

MYUNG GUK LEE

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US CITIZEN

Portfolio : <https://myung.onrender.com>

WORK EXPERIENCE

Innophase • Irvine, CA

03/2023 - Present

Staff Firmware Engineer

- Designed and implemented a UART-based daisy-chain communication system on Azure RTOS, enabling coordinated serial communication across up to N devices.
- Implemented a JSON decoder library with static memory allocation for memory-constrained embedded systems, ensuring predictable memory usage and validating reliability through unit testing.
- Developed an SPI/QSPI NOR flash device driver integrated with FileX/LevelX on Azure RTOS.
- Translated MATLAB-based Digital Pre-Distortion(DPD) algorithm into production C firmware, implementing complex mathematical models (matrix operations, polynomial computations, iterative convergence) for real-time signal processing on ARM Cortex-A microcontrollers under strict memory and timing constraints.
- Optimized DPD algorithm performance by 3x through ARM NEON SIMD vectorization.
- Brought up PCIe on a TI AM64x EVM in bare-metal firmware, implementing controller initialization, configuration-space enumeration, and DMA-based data transfer to validate end-to-end PCIe behavior on a new SoC. Created the internal PCIe reference implementation for RTL engineers.

Western Digital Inc. • Irvine, CA

09/2018 - 03/2023

Principal Firmware Engineer

- Integrated TCG, ATA Security, and Sanitize features - including on-media data encryption (SED), into enterprise HDD firmware (SATA/SAS) across multiple customer products that shipped to production; resolved security-firmware defects through qualification and release.

Staff Firmware Engineer

06/2014 - 09/2018

- Developed a PCIe AHCI device driver for hybrid HDDs, enabling SSD-tier caching and improving performance.

SKILLS

- Language

- C/C++ , Python

- Networking

- TCP, UDP, ARP, DHCP, VLAN, L2/L3 Ethernet switches

- Embedded OS

- Azure RTOS(ThreadX), Embedded Linux, FreeRTOS, Bare-Metal Firmware

- HW Interfaces

- SPI/QSPI, UART, GPIO, PCIe, SATA/SAS

- Tools & Debug

- JTAG , Oscilloscope, Git

- Security

- TCG/ATA Security, On-Media Encryption - (SED)

- ARM NEON SIMD

EDUCATION

MS in Computer Science

University of California, Irvine, CA

BS in Computer Science

Hanshin University

- Developed DUAL-SIO, a half- and full-duplex serial protocol, improving test equipment flexibility and reducing manufacturing costs.

AnyData •Irvine, CA

06/2012 - 06/2014

Senior Firmware Engineer

- Implemented Firmware Over-the-Air (FOTA) upgrade support, enabling remote firmware updates and improving product maintainability.

DASAN Networks • South Korea

12/2005 - 06/2010

Software Engineer

- Led embedded software development for L2 Ethernet switches, implementing complex networking protocols and managing real-time packet processing using C/C++ on Linux systems with Broadcom and Marvell ASICs.
- Implemented a high-performance packet monitoring system using the TCP/IP stack, Unix domain sockets, and shared memory IPC to support real-time troubleshooting for enterprise L2 switches.
- Developed network security features, including DHCP snooping and ARP inspection, to improve switch-level protection and traffic integrity.
- Developed a subscriber access monitoring system for Korean Telecom (KT), improving network management efficiency and service reliability.

AWARDS

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- **Spot Award Winner • Innophase• 2024**
 - **High-Five employee recognition • WesternDigital• 2018**
 - **Best Thesis •Korean Information Processing Society• 2012**
 - **Summer Code Competition •ICS of UC,Irvine• 2012**